Linear Algebra — Concept Map
how we built from idea to idea

study notes

August 29, 2026

1 The mind map

2 How we built: concept → concept

1. **Vector space** — the foundation. A set with addition and scalar multiplication (closure, zero vector, axioms). A vector is any element; in $\mathbb{R}^n$ it's a stack of numbers.
2. **Representation → dimension.** $\mathbb{R}^n$ has n directions, P_n has $n+1$, P has $\aleph_0$, C is uncountable. Dimension = number of independent directions (basis size), not vector length.

3. **Free vs fixed.** Displacement vs position reading; the math runs on displacements (a basis, never an origin). This is why $[3, 5]$ needs no reference point as a displacement.

4. **Norm.** Add "size" to a vector space (3 axioms: positive definiteness, homogeneity, triangle inequality). Different norms = different geometry (unit spheres: circle, diamond, square, ellipse).

5. **Metric / distance.** Every norm induces a metric $d(x, y) = \|x - y\|$; "$\|x\|$ = distance to the origin" is the position-reading corollary, not the definition.

6. **Triangle inequality.** Its engine is the projection split: $v = v_\parallel + v_\perp$, and $\left\|v_\parallel\right\| \le \|v\|$ — a projection is never longer than the vector it came from.

7. **Cauchy–Schwarz.** That engine in dot-product clothes: $|\langle u, v\rangle| \le \|u\| \, \|v\|$. Same fact, two languages.

8. **Inner product.** Add "alignment" between two vectors ($\langle x, y\rangle = x^T y$). It induces the norm $\|x\| = \sqrt{\langle x, x\rangle}$ and brings angles, orthogonality, and projections.

9. **Parallelogram law / polarization.** The bridge back: a norm comes from an inner product iff $\|u + v\|^2 + \|u - v\|^2 = 2(\|u\|^2 + \|v\|^2)$ (Jordan–von Neumann), and polarization recovers the inner product from the norm — the norm was the inner product in disguise.

10. **Weighted inner products.** Any symmetric positive-definite M works ($\langle x, y\rangle_M = x^T M y$); the M norm is one of them. ℓ_2 is the flagship ($M = I$), the only rotation-invariant norm. That's the deep payoff: *the dot product induces the L2 norm, and L2 is the only norm invariant under rotation* — which is why L2's geometry matches everyday space, while the weighted norms are just L2 recoordinatized.

3 The ladder of structure

Every step adds structure; the richest sits at the top.

$$\begin{array}{rcl} \text{inner product space} & \subset & \text{normed space} \subset \text{metric space} \\ \text{(sizes + angles)} & & \text{(sizes)} \quad \text{(just distances)} \\ \text{vector space is the base: set + addition + scaling} & & \end{array}$$

- **vector space → normed space**: add $\|\cdot\|$ (3 axioms).
- **normed space → inner product space**: add $\langle\cdot, \cdot\rangle$, which induces the norm.
- **normed space → metric space**: every norm gives $d(x, y) = \|x - y\|$ (the triangle inequality is the norm's own axiom).
- Each step up is one-directional: inner product ⇒ norm ⇒ metric, but not back (except when the parallelogram law holds).